

# Autonomous Agent-aware Navigation in Underground Mine Environments

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**Abstract**— This paper discusses the challenges associated with the navigation of robots in mine environments and the need for advanced robotic systems that can operate effectively and safely in these environments. The paper highlights the importance of human-aware navigation and the compliance with safety regulations in mine environments. The paper proposes a motion model for tracking agents and proxemic-influenced behaviors to increase the robots’ behavior consistency. Intersection detection and landmark-based localization are also employed to improve safe navigation.

## I. INTRODUCTION

Mines are known to be hazardous environments for human workers, and accidents and fatalities can occur due to various factors such as rock falls, equipment failures, explosions, and exposure to hazardous gases. According the U.S. Department of Labor, there were 29 mine-related deaths in the U.S. in 2020 [4]. Additionally, there are long-term health risks associated with mining, such as respiratory diseases and cancer due to exposure to dust and other harmful substances.

Given the dangers associated with mining, the use of robots in these environments presents an opportunity to improve safety and reduce the risk of human injury or fatalities. By replacing human workers with robots, the risk of accidents and fatalities can be greatly reduced. Moreover, robots can be used to access and explore areas that are too dangerous for humans to enter, such as areas with high levels of dangerous gases or radiation. This not only improves safety but also allows for more efficient and effective exploration and mining operations.

Navigating in a mine environment presents significant challenges for robotic systems due to factors such as limited visibility, narrow corridors, lack of GPS, limited communication, and a resource-constrained environment. In these types of environments, there is a lack of any natural illumination, unconditioned narrow passages, crossing paths, dirt, high moisture and dust [3]. These challenges can have a significant impact on the safety and efficiency of robotic operations in mines. As a result, there is a growing need for advanced robotic systems that can navigate and operate effectively in these environments.

Despite the desire to deploy robots in mines to minimize human exposure to dangerous conditions, it is likely



Fig. 1. Example robot-camera view of the simulated mine environment.

that humans will still need to work in these environments, and robots will need to operate in a way that takes humans’ presence into account to ensure safe and efficient co-operation. Robot navigation in mines that are actively being worked by humans introduces additional challenges. Human safety must be prioritized, and robots must be able to navigate in a way that minimizes the risk of collisions and other safety hazards. This requires human-aware navigation, which involves detecting and tracking human workers and machinery in real-time and using this information to adjust robot trajectories and behaviors accordingly. Human-aware navigation is especially important in mines where humans are actively working, as humans and human-operated machinery will not prioritize robot navigation over their own tasks, and collisions between robots and humans can have serious consequences. Therefore, developing robotic systems that can safely navigate and operate in mines while being aware of the presence of human workers is a critical challenge that must be addressed in this field.

In addition, strict safety regulations often exist in mine environments. These regulations are put in place to protect human workers and ensure safe operations in the mine. As a result, robots operating in these environments must comply with these regulations, which can significantly impact their navigation strategies. For example, in some mines, agents are required to “honk” before entering an intersection to alert other workers or robots to their presence. Failure to do so can result in collisions and other safety hazards. Additionally, speed limits and restrictions on certain areas of the mine may also exist, which must be adhered to by the robotic systems. Compliance with these regulations requires the development of advanced sensing and perception capabilities to detect and interpret the relevant cues and signals, as well as the ability to adjust robot behaviors in real-time to adhere to

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these regulations.

Human-aware navigation is an active research field. For example, previous efforts have shown how to dynamically select between three closed loop policies that best fits the situation [1], and employ deep reinforcement learning to achieve collision avoidance in person-rich environments [2]. This work seeks to solve similar problems subject to the additional constraints and challenges of mine environments. To address these challenges, We present initial efforts in the design of a human-aware navigation system to compute motion prediction models for dynamic agents (human and machine) and use available signage to aid localization in a mine environment.

## II. PROBLEM STATEMENT

The problem addressed in this research is the safe and efficient navigation of a mobile robot in a mine populated by both the robot and dynamic agents. The robot must navigate through long orthogonal drifts, deep underground and with limited resources, all while avoiding collisions with other agents in the environment.

It is assumed the maximum speed allowed for all agents is  $4.5m/s$ , the passageways in the mine are parallel or orthogonal to each other, refer to Fig. 1, and the environment as a whole is not feature rich. This combination leads to issues in navigation and localization for the robot. The mine environment contains signage that is unique to specific locations, such as intersections labeled with mine coordinates. While this information can be used to aid in navigation, it adds another layer of complexity to the problem.

This research aims to address this problem by developing a navigation strategy that takes into account the unique challenges of the mine environment and the presence of a multitude of agents.

## III. APPROACH

Our approach involves computing a motion model for each dynamic object, either a person or a vehicle, we will refer to them as tracking agents, to allow for more accurate navigation in the environment, proxemic-influenced behaviors to increase the robots behavior consistency, intersection detection, and landmark-based localization.

In our formulation, time-horizon refers to the number of seconds a motion model of a tracking agent is predicted into the future. Position history refers to the number of seconds in the past over which position data is collected of a tracking agent and used to calculate a motion model. Motion prediction refers to the process of estimating the future motion or trajectory of an object or agent based on its current state and past observations.

The goal of motion prediction, in our case, is to anticipate the future behavior of a tracking agent in order to make informed decisions. Our initial approach uses a simple second degree polynomial to model the tracking agent's motion,  $P(t)$ . Using position data over the position history, a polynomial fitting algorithm is used to calculate the coefficients. The input is the time in seconds the position was observed and the

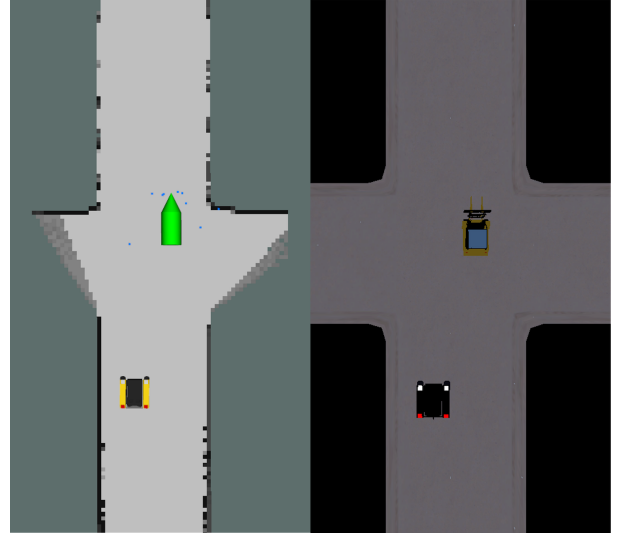


Fig. 2. Side-by-side of the simulator and Rviz view of a robot observing a forklift driving away. The green arrow is representative of its current predicted motion.

output is a set of coordinates,  $(x, y)$ . Points are then generated with this polynomial by getting coordinates between current time and the time-horizon in one second intervals. Each point is then used as the input for a normal Gaussian distribution with a  $\sigma$  of 0.9 generating 10 points each. These points,  $F_{predicted}$ , are meant to characterize the potential motion of the tracking agent over the time-horizon. Fig. 2 shows the motion prediction for a forklift in the simulated environment. These points are then projected onto the global occupancy grid which is used to generate a global plan that the robot will follow to reach its end goal. If the projected motion of the dynamic agent intersects with the global plan between current time and the time-horizon, the robot re-plans its global plan. At any point in time, if the distance between the robot and a dynamic agent is below a specified threshold, the robots priority is to slow, move to the right side of the passage-way, and stop.

We approach detecting intersections in the mine environment through the use of sensors such as LiDAR, which can detect changes in the geometry of the environment. By analyzing the data collected by these sensors, we can identify locations where multiple passageways intersect, indicating the presence of an intersection.

Once intersections have been detected, we can use the signage in the mine as landmarks for navigation. By comparing the coordinates on the signs with the robot's internal map, the robot can determine its location relative to the intersection and use this information to navigate to its destination. This LiDAR-based SLAM with landmark-based localization approach allows for accurate localization and navigation in the GPS-denied environment.

Overall, motion prediction, intersection detection, and signage for landmark-based localization, along with visual odometry techniques, presents a promising approach for safe and reliable robot navigation in a human worker populated mine environment.

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